

1

Ademy Course
ML, NLP, DL Course
Krush Naik sir
Sem - 22

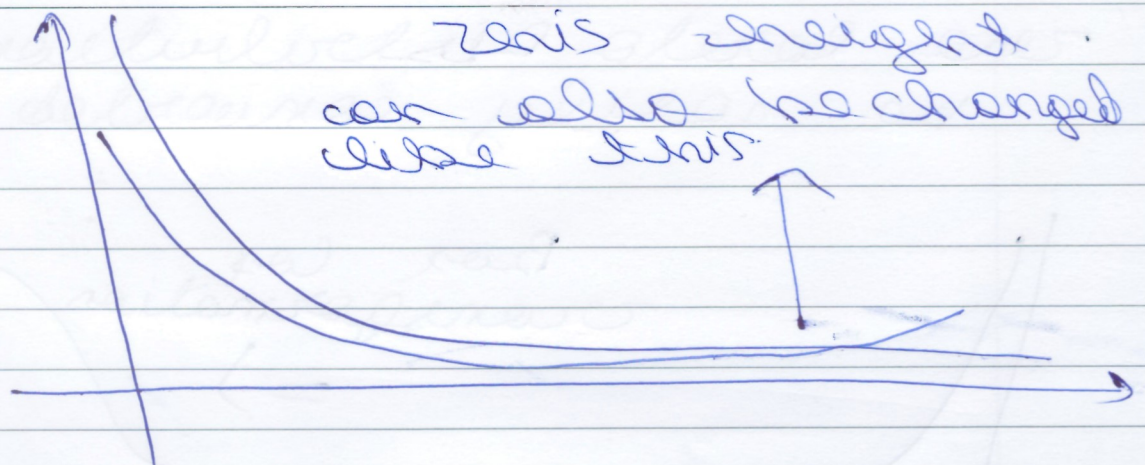
Probability Distribution
Frame for all.

Lec - 11 Poisson
Distribution

Lec - 12 Central Limit
Theorem

Lec - 13 Estimator

Poisson Distribution.
[Non-Gaussian
Distribution]

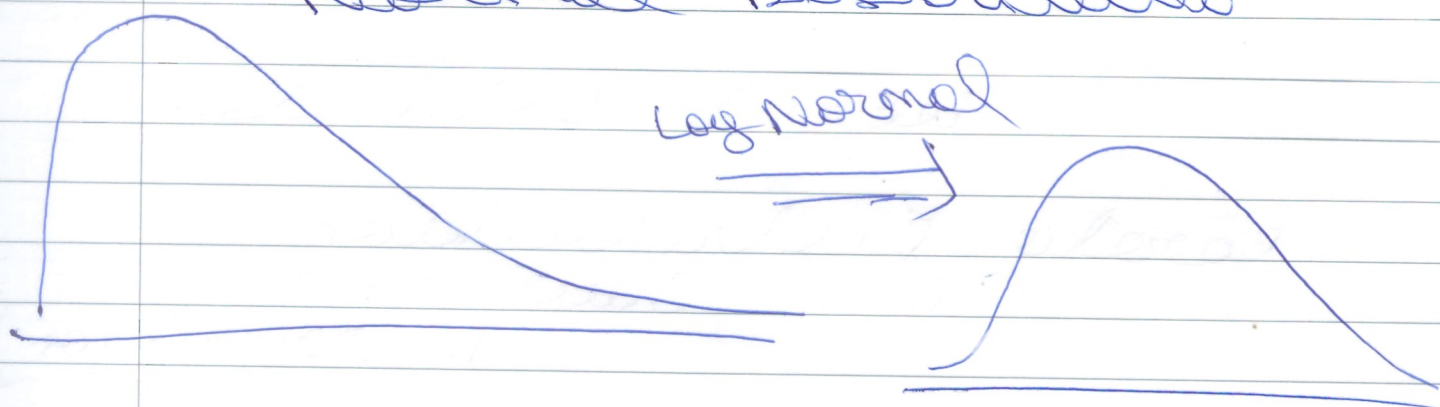


In the case of Pareto Distribution, it is mainly defined by α

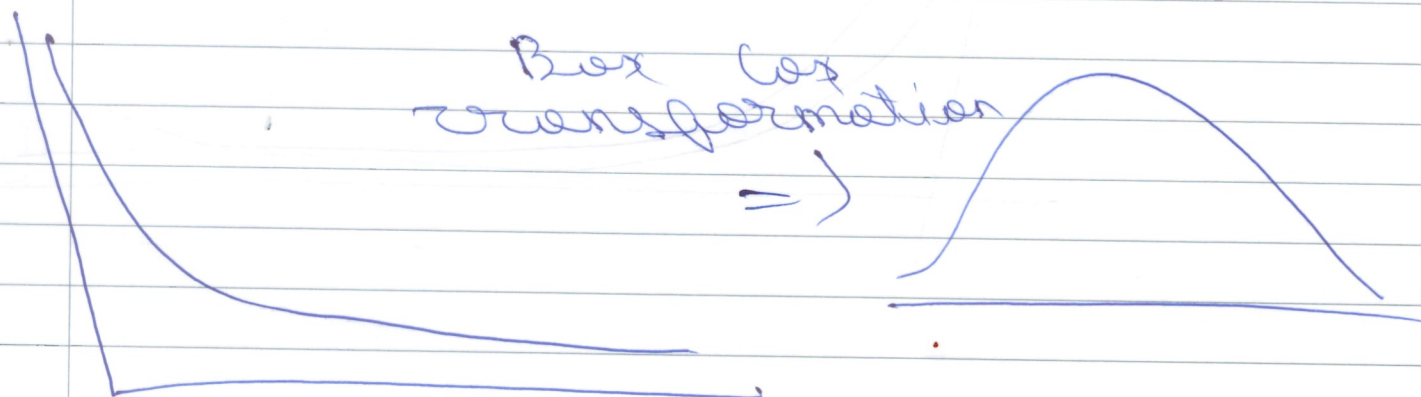
α value keeps on increasing.

It will be reaching further and further to this line of 1.

Suppose we have Log Normal distribution



The Pareto Distribution can be easily converted to



Expt 1 IT Industry

(1) 80% of the entire project is done by 20% of team is known as 20% of the team.

(2) 20% of ~~the~~ effort can be achieved if we use 20% of the team.

Power Law Distribution follows the power law distribution.

lec - 12

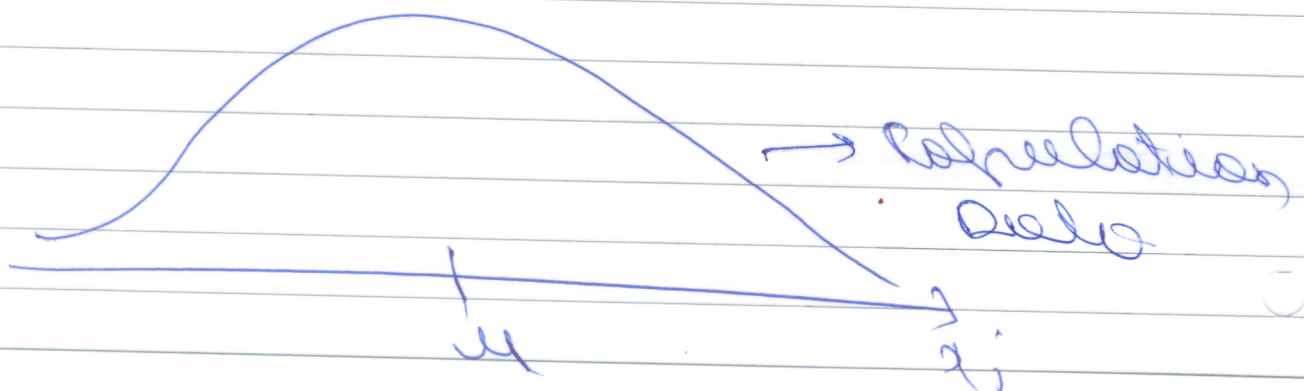
Central Limit Theorem

Central Limit Theorem
relates the concept of
sampling distribution,
which is probability
distribution for a large
no. of samples from
a population.

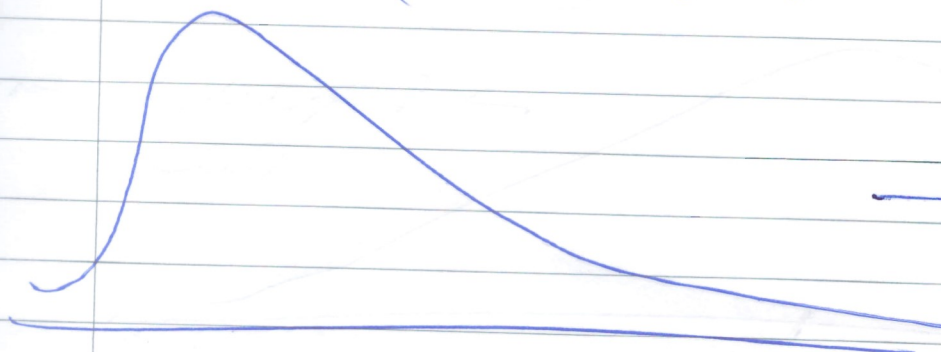
The central limit
theorem says that the
sampling distribution of
sample means will always be
normally distributed, as
long as sample size is
large enough.

(1)

$$X \sim N(4, \sigma)$$



② $X \sim N(\mu, \sigma)$



→ This is the right skewed or left skewed.

$n \geq 30$

→ sample size.

All this sample will also follow Gaussian distribution.

s_1

$\bar{x}_1$

$\bar{x}_2$

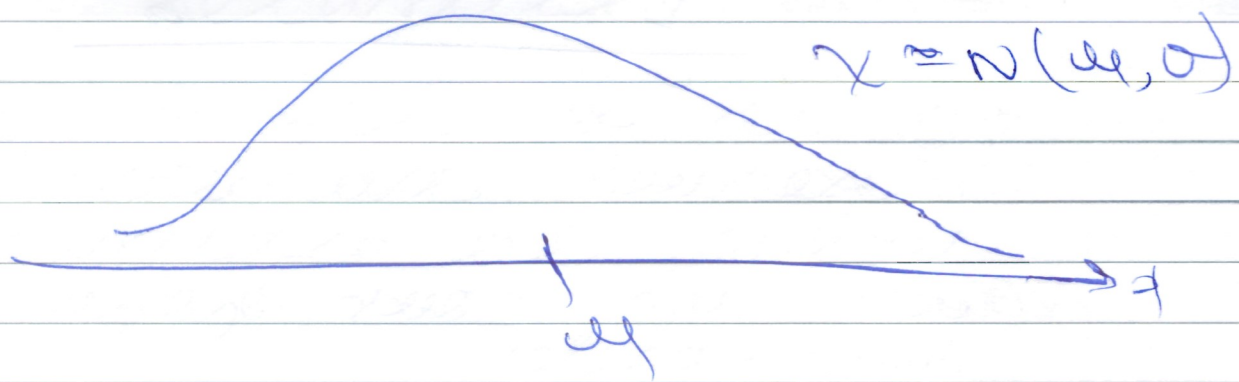
s_2

— — — — —

$\bar{x}_n$

7

Normal Distribution



Sampling Distribution
of mean



μ = Population mean

σ = Population Standard Deviation

n = sample size

lec - 13

Estimators

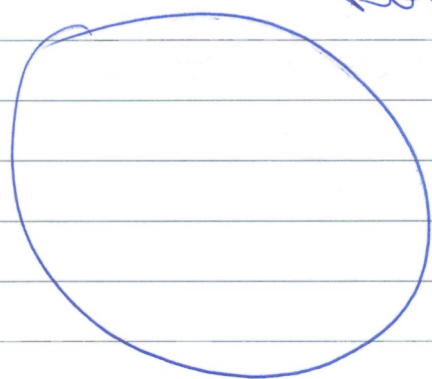
It is a specified numerical value or estimator used to estimate an unknown population parameter.

Different types of estimators

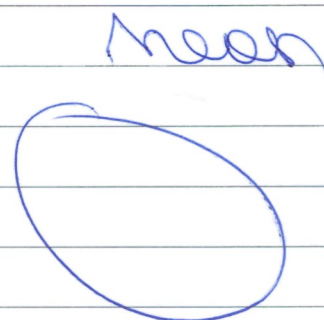
(1) Point Estimator:-

single numerical value used to estimate unknown population parameter.

Ex:- Sample mean is a point estimator of population mean.



μ Population
(sample mean)



$\bar{x}$ Mean
(sample mean)

② Interval Estimate :-

- Range of values is used to estimate unknown population parameter.

→ Used to specify range

We can see range in SS - 65

